DATA ANALYTICS TASKS 2-4

Solution Report By Muhammad Umer

Tasks 2, 3 and 4

Coverage: data integrity and exploratory analysis, customer segmentation and cohort dashboard, inferential statistics and business forecasting.

IMPORTANT DATA LIMITATION: The supplied instruction PDF contained task descriptions only and did not include the original industry dataset. To provide a runnable end-to-end solution, the technical examples use a reproducible synthetic retail dataset. All numbers and findings in this report are illustrative and must not be presented as real company results.

Executive summary

Metric	Demo result
Raw records	20,200
Clean records retained	19,649
Unique customers	2,412
Total revenue in demo data	3,283,917.72
Top demo category by revenue	Technology
Regression model	RandomForestRegressor (test R-squared 0.878)
Forecast	ARIMA(1,1,1), 30-day horizon

Deliverables included with the ZIP

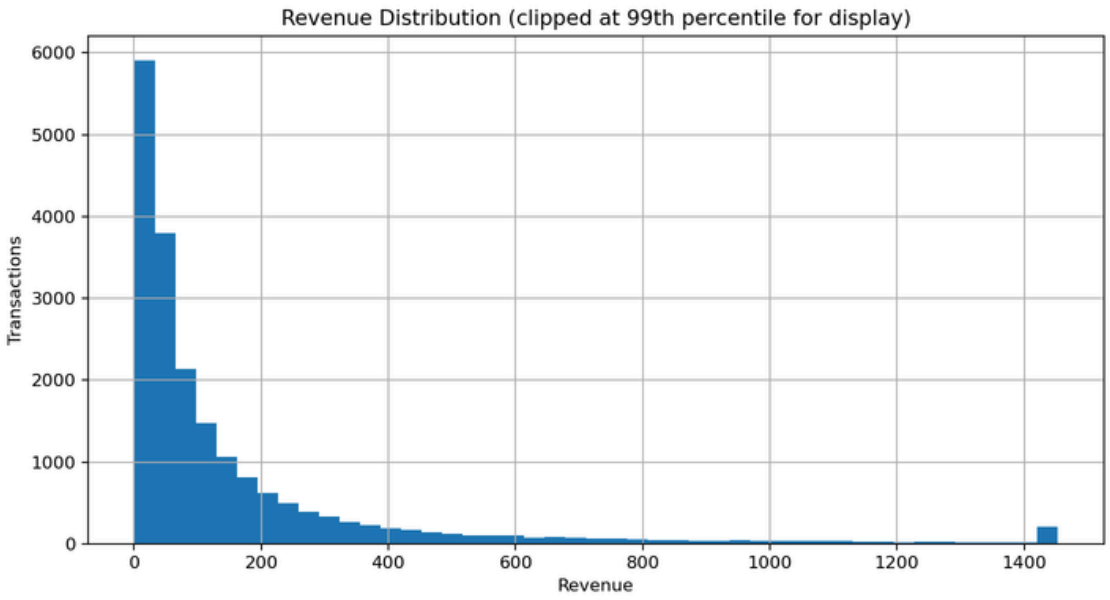
Runnable Jupyter notebook, Python analysis pipeline, demo data generator, Streamlit dashboard application, Plotly HTML dashboard, cleaning diagnostics, RFM and cohort CSVs, hypothesis-test outputs, regression evaluation, and forecast intervals.

Task 2 - Exploratory Data Analysis and Integrity Cleaning

Data qualitycheck	Count
Raw rows	20,200
Exact duplicates detected and removed	200
Invalid dates after parsing	80
Rows dropped for invalid dates	80
Rows dropped for missing customer IDs	239
Rows dropped for impossible quantity/price	32
Rows affected by IQR outlier capping	691
Clean rows retained	19,649

Cleaning policy

The pipeline removes exact duplicate rows; parses mixed date formats; drops invalid dates instead of inventing timestamps; excludes impossible non-positive quantity/price records and rows without customer IDs; imputes numeric missing values by product-category median (with global median fallback); imputes categorical values by mode; and caps Quantity and UnitPrice using IQR fences. It derives GrossSales from quantity, price and discount for reconciliation, while preferring a provided ReportedRevenue column as the analysis outcome. This is a sample policy and must be revised for real data, particularly where negative quantities represent valid refunds.

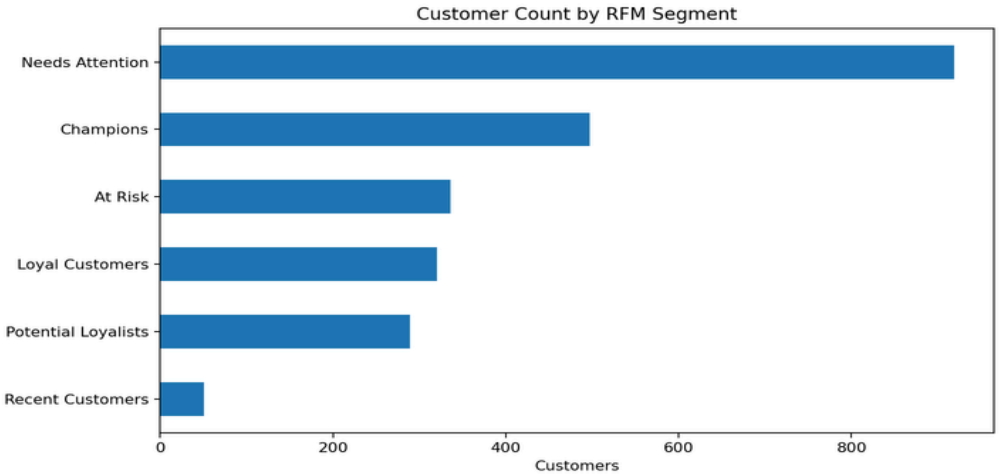


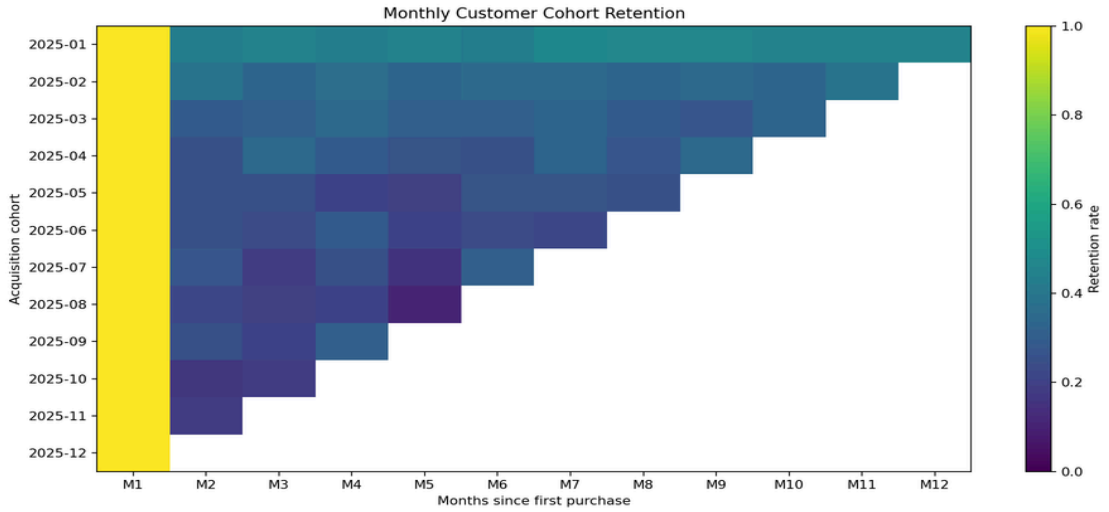
Task 3 - Customer Segmentation and Behavioral Trends

RFM definitions

Recency = days since the customer's last observed purchase; Frequency = distinct invoices; Monetary = total revenue. Scores from 1 to 5 are assigned using quantile buckets. Rule-based segments include Champions, Loyal Customers, Potential Loyalists, At Risk, Recent Customers, and Needs Attention.

RFM segment	Customers	Total demo revenue
Needs Attention	919	371,159
Champions	497	2,048,222
At Risk	336	316,705
Loyal Customers	320	357,961
Potential Loyalists	289	177,701
Recent Customers	51	12,169





The Streamlit dashboard (app.py) supports date, category, country, campaign and RFM segment filters; shows daily revenue, category contribution, customer segments and a country revenue map; and allows exporting filtered transaction and RFM CSV files. The CSV exports can also be imported into Power BI or Tableau.

Task 4 - Inferential Statistics and Predictive Forecasting

Hypothesis tests

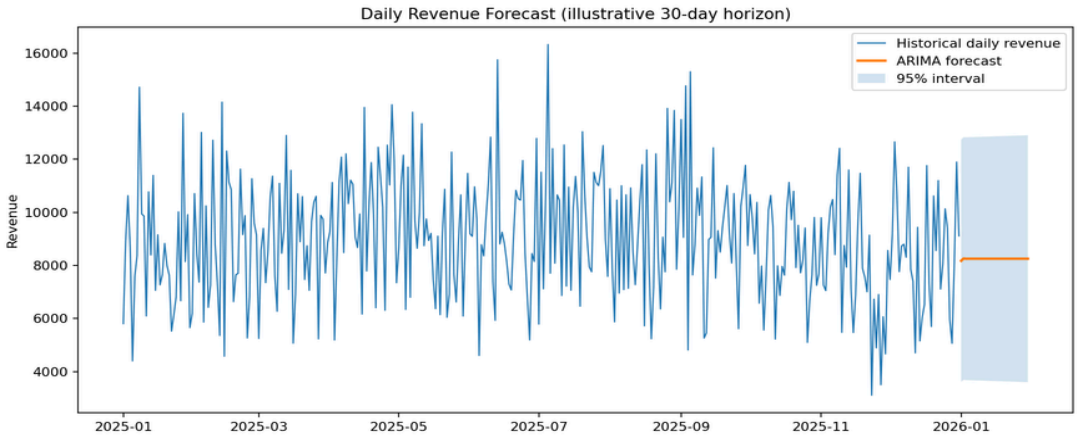
Test	Statistic	p-value	Interpretation at alpha = 0.05
Welch t-test: Campaign B vs A mean Revenue	3.585	0.000338	Statistically significant difference
One-way ANOVA: Revenue by ProductCategory	2661.912	<1e-300*	At least one category mean differs
Chi-square: CampaignGroup vs Returned	23.694	1.13e-06	Evidence of association

*A zero p-value in this output reflects floating-point underflow for an extremely small computed p-value, not a literally zero probability.

The Welch t-test compares transaction revenue in demo Campaign A vs B, ANOVA checks whether mean revenue differs across product categories, and chi-square tests association between campaign group and return status. Statistical significance does not establish causality or business value. The synthetic generator deliberately adds a campaign uplift and different return probabilities, so these results are for method demonstration only.

Regression and forecast evaluation

Evaluation	Result
Regression model	RandomForestRegressor
Target	Revenue
Test rows	3,930
Test R-squared	0.878
Test MAE	35.47
Test RMSE	98.12
Forecast model	ARIMA(1,1,1)
Forecast dates	2026-01-01 to 2026-01-30
Mean forecast daily revenue	8,251.74



Limitations: benchmark the regression against simple baselines and check for leakage; evaluate forecasting against seasonal-naive and business baselines; revisit temporal split and seasonality assumptions; and validate test assumptions and practical effect sizes using real data.

Reproducibility

From the project folder, install requirements.txt, run `python src/analysis_pipeline.py --data data/retail_transactions_demo.csv --output outputs`, then run `streamlit run app.py`. The notebook contains the end-to-end walkthrough and displays generated tables and charts.